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Improved Turbocharger Compressor (Cold Side) Oil Leaks Troubleshooting

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document introduces new troubleshooting for QSK19 engines that have oil leaks into the engine intake side of the turbocharger.

Reference the Turbocharger Leaks Engine Oil or Fuel troubleshooting symptom tree in Section TS in the QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS Service Manual, Bulletin 4021592.

The new troubleshooting tree outlines steps that must be taken before the turbocharger can be replaced. This is due to a high number of turbocharger replacements that were done on turbochargers that were found to be in working mechanical condition.

Oil leaks from the compressor (cold) side of the turbocharger do not always indicate a malfunction of the turbocharger oil seal. The oil seal is a piston-ring type seal, (see Figure 1) that is designed to function under a positive pressure differential between the compressor backwheel pressure (related to boost pressure) and the turbocharger bearing housing pressure (related to crankcase pressure).

If the compressor backwheel pressure drops below the bearing housing pressure, oil can be forced past the seal and into the compressor housing of the turbocharger. If bearing wear/damage has not occurred (radial and axial end play is still in specification), the turbocharger itself will be undamaged. It can be cleaned and put back into service. For cleaning instructions, Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html>)



Figure 1 - Turbocharger Compressor Oil Seal

This failure mode is often identified by staining, oil pooled in the bottom of the compressor housing, or oil found in the intake piping downstream from the turbocharger. Often, the root cause of the problem can be related to low boost pressures or elevated bearing housing pressures, and not a malfunction of the turbocharger oil seal.

If oil leaks are seen in this area, follow the troubleshooting procedure linked above before replacing the turbocharger.

Document History

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